

W-6 Nowcasting Models with Non-Traditional Data

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Review of Previous Nowcasting Model

Without Non-Traditional Data

Equation: EQ01_F_BRIDGE Workfile: NWC::NOWCAST_RE...

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: DLOG(RGDP)
Method: Least Squares
Date: 12/19/24 Time: 10:34
Sample (adjusted): 2000Q2 2023Q1
Included observations: 92 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(RGDP(-3))	-0.017240	0.038557	-0.447114	0.6560
DLOG(RGDP(-2))	-0.040927	0.040348	-1.014354	0.3134
DLOG(RGDP(-1))	-0.032954	0.065557	-0.502680	0.6165
DLOG(CONSU_MOTOR)	0.312822	0.036000	8.689424	0.0000
D(PMI_NEW)	0.001671	0.000400	4.180365	0.0001
DLOG(RGDP_US(-1))	0.091918	0.166480	0.552128	0.5823
DLOG(RGDP_CN(-1))	0.376061	0.121011	3.107649	0.0026
CRISIS	-0.022421	0.007314	-3.065602	0.0029
C	0.003694	0.002779	1.329025	0.1875

R-squared	0.926203	Mean dependent var	0.015577
Adjusted R-squared	0.919090	S.D. dependent var	0.040440
S.E. of regression	0.011503	Akaike info criterion	-5.999733
Sum squared resid	0.010982	Schwarz criterion	-5.753036
Log likelihood	284.9877	Hannan-Quinn criter.	-5.900164
F-statistic	130.2139	Durbin-Watson stat	2.221487
Prob(F-statistic)	0.000000		

Evaluation statistics

Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
BRIDGE_3	1044.619	678.2968	1.973451	1.937896	0.014282	0.364241
MIDAS_3	1433.933	826.5685	2.454167	2.400210	0.019608	0.518452
UMIDAS_3	1741.772	1024.135	3.030600	2.966029	0.023932	0.629249
FC_3	1279.524	745.7753	2.216474	2.162386	0.017525	0.449567

Note: this is out-of-sample forecast evaluation at quarter-end

Google Trend Data

India Tourism Keywords

Table: UNTITLED Workfile: NWC::Quarterly\

View | Proc | Object | Print | Name | Edit+/- | CellFmt | Grid+/- | Title | Comments+/-

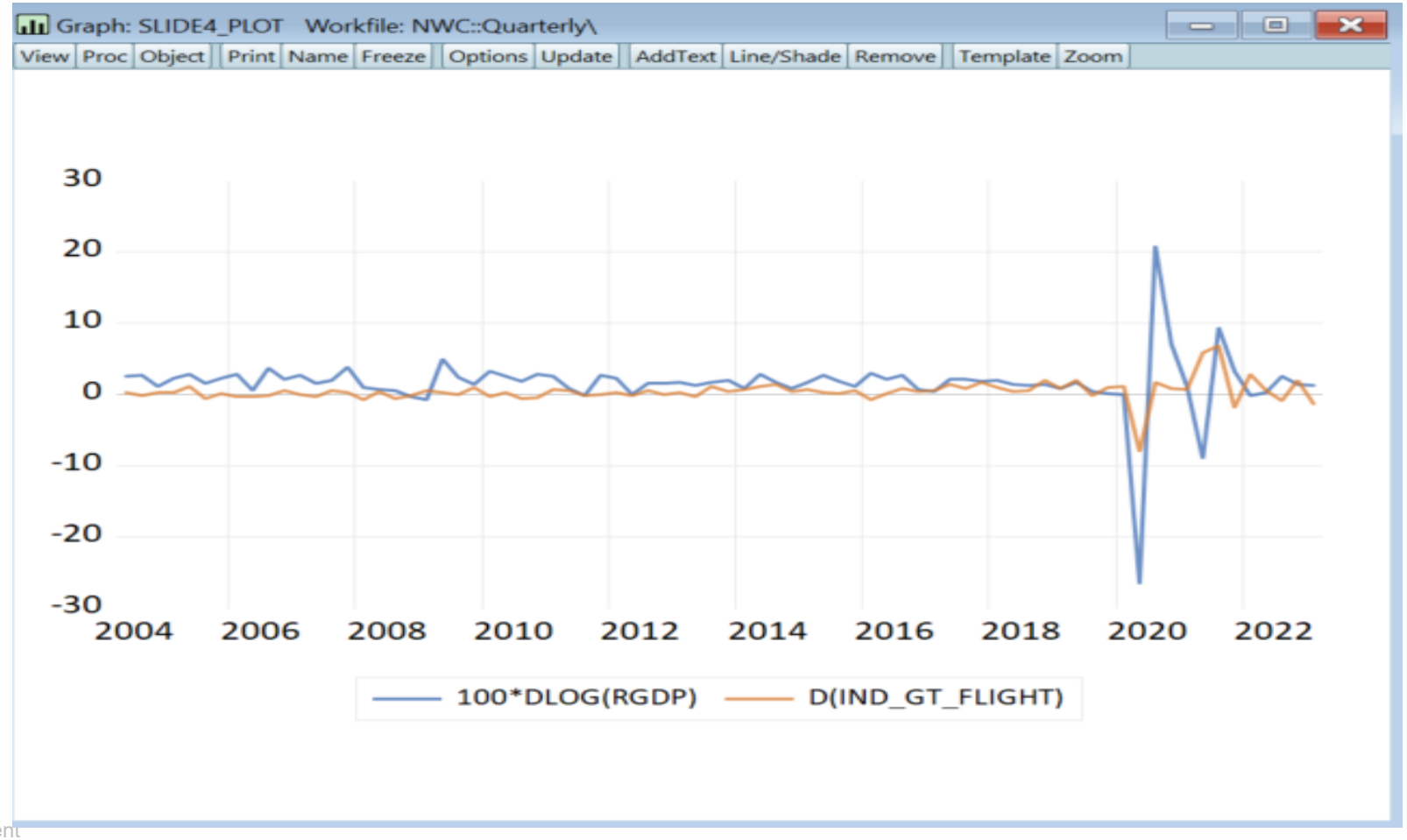
Date: 01/03/25 Time: 12:32
Sample: 2004Q2 2023Q1
Included observations: 76
Balanced sample (listwise missing value deletion)

Covariance / Correlation	DLOG(RGDP)	D(IND_GT_FLIGHT)	D(IND_GT_HOTEL)	D(IND_GT_TM)	D(IND_GT_TRAVEL)
DLOG(RGDP)	0.001916				
	1.000000				
D(IND_GT_FLIGH...	0.096605	22.12745			
	0.469150	1.000000			
D(IND_GT_HOTEL)	0.082817	9.104763	26.15573		
	0.369925	0.378460	1.000000		
D(IND_GT_TM)	0.034799	10.09153	26.99322	306.7943	
	0.045386	0.122481	0.301334	1.000000	
D(IND_GT_TRAV...	0.064727	26.49492	0.921009	-17.16945	85.12295
	0.160266	0.610483	0.019519	-0.106245	1.000000

Google Trend Data
India Tourism Keywords (Hotel)

Real GDP Growth Rate vs. Google Trend (India Flight)

Absolute Correlation: 0.47 (2004Q2~2023Q1)



Update Previous Nowcasting Model
With Non-Traditional Data (Google Trend)

Equation: EQ01_F_BRIDGE Workfile: NWC::Bridge\

View | Proc | Object | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Dependent Variable: DLOG(RGDP)
Method: Least Squares
Date: 12/19/24 Time: 11:08
Sample (adjusted): 2004Q2 2023Q1
Included observations: 76 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(RGDP(-3))	-0.025050	0.037600	-0.666210	0.5076
DLOG(RGDP(-2))	-0.053805	0.042953	-1.252648	0.2148
DLOG(RGDP(-1))	-0.010452	0.067450	-0.154963	0.8773
DLOG(CONSU_MOTOR)	0.318402	0.037406	8.511950	0.0000
D(PMI_NEW)	0.001515	0.000372	4.067407	0.0001
D(IND_GT_FLIGHT)	0.000591	0.000421	1.403905	0.1650
DLOG(RGDP_US(-1))	-0.009726	0.201140	-0.048353	0.9616
DLOG(RGDP_CN(-1))	0.307728	0.120674	2.550067	0.0131
CRISIS	-0.026076	0.007185	-3.629137	0.0006
C	0.005721	0.002790	2.050940	0.0442

R-squared	0.949888	Mean dependent var	0.016302
Adjusted R-squared	0.943055	S.D. dependent var	0.044065
S.E. of regression	0.010515	Akaike info criterion	-6.149869
Sum squared resid	0.007298	Schwarz criterion	-5.843193
Log likelihood	243.6950	Hannan-Quinn criter.	-6.027306
F-statistic	139.0060	Durbin-Watson stat	2.357172
Prob(F-statistic)	0.000000		

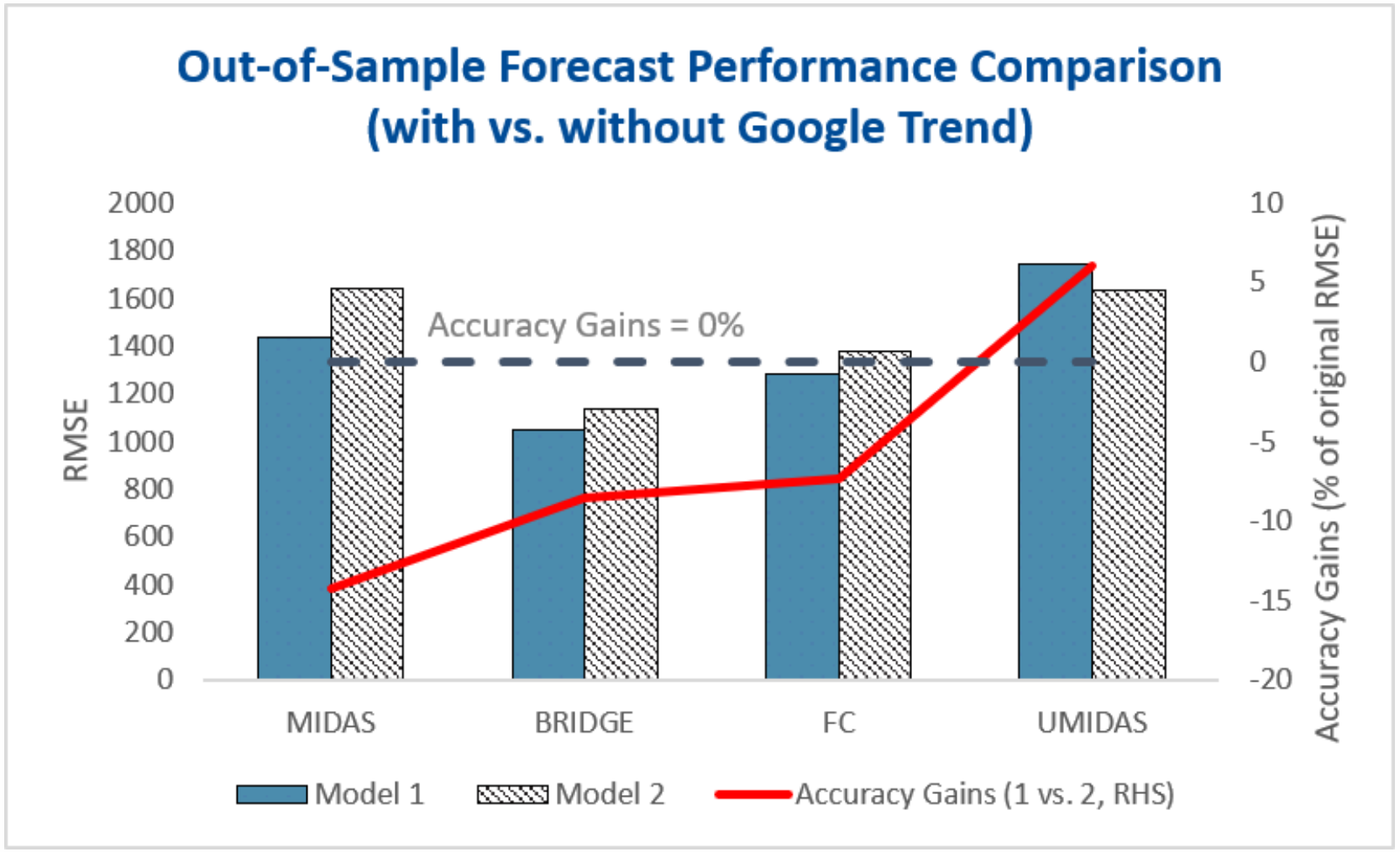
Evaluation statistics

Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
BRIDGE_3	1134.592	733.2810	2.138472	2.099892	0.015501	0.387813
MIDAS_3	1638.272	935.5842	2.817124	2.728472	0.022378	0.588154
UMIDAS_3	1636.204	1096.176	3.195517	3.157722	0.022448	0.588977
FC_3	1374.052	847.5810	2.516534	2.458163	0.018799	0.483533

Note: this is out-of-sample forecast evaluation at quarter-end

Out-of-Sample Forecast Performance Comparison

Original Model vs. Model with Google Trend



Note:
Model 1: Original Model
Model 2: Original Model + Google Trend

**Note: Accuracy Gains in % of original RMSE in the bracket, across 2018Q1 and 2022Q3*

Summary

Usefulness of Big Data and Important Remarks

Adding Google Trend only improves out-of-sample forecast performance for UMIDAS and UMIDAS-PC models, along with selected India baseline model.

3 Important Conditions for Regressors

1. **Coefficients**
   The sign of regressors' estimated coefficients should be consistent with standard economic prior
2. **Stationarity**
   All regressors should not possess a unit root
3. **No Seasonality**
   Add appropriate dummies to the baseline regression to correct for non-seasonality or seasonally adjust the series using X-13 procedure

3 Criteria for Model Selections

1. **Akaike Info Criterion**
   Used in forecasting/nowcasting model selection. Smaller values are preferred
2. **Adjusted R-Squared**
   Adjust for the number of independent regressors by penalizing R-Squared for additional regressors. Larger values are preferred.
3. **Root Mean Squared Error (RMSE)**
   Frequently used measure of the differences between values predicted by a model and the values observed. Smaller values are preferred.

Thank You!